ADJACENT-RUN CONSOLIDATION OF INTEGER COMPOSITIONS: A SHARP LOGARITHMIC CLOCK AND DIVISOR-PATH FIBRES

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Abstract. Fix a positive integer n. In a composition of n, replace every maximal run of r equal parts s simultaneously by the single part rs, and iterate. This weight-preserving map is absorbed at the adjacent-unequal (Carlitz) compositions. We prove that its largest absorption time is exactly $\lfloor \log_2 n \rfloor$ for every n, and give an explicit equality witness for every carrier size. We also determine every one-step fibre, refined by source length: for a target $\beta = (b_1, \ldots, b_k)$, its fibre polynomial is the sum of $u^{\sum_i b_i/s_i}$ over divisor choices $s_i \mid b_i$ satisfying $s_i \neq s_{i+1}$. Thus the inverse is a weighted path problem on adjacent divisor sets. An exhaustive exact-arithmetic falsifier checks all 262,143 compositions of total at most 18 and performs 2,690,869 assertions. Classical Carlitz enumeration and static run statistics are treated as background, not as contributions. The note is anonymous and remains under `HOLD_EXTERNAL`.

1. The finite map and the theorem

Let $\mathrm{Comp}(n)$ be the set of ordered tuples of positive integers summing to n. Every $\alpha \in \mathrm{Comp}(n)$ has a unique maximal-run factorisation

$$\alpha = s_1^{r_1} s_2^{r_2} \cdots s_k^{r_k}, \qquad s_i \neq s_{i+1},$$

where s^r means r adjacent copies of s. Define

$$A_n(\alpha) = (r_1 s_1, r_2 s_2, \ldots, r_k s_k). \tag{1}$$

The update is simultaneous. For example,

$$(1, 1, 2, 4) \longmapsto (2, 2, 4) \longmapsto (4, 4) \longmapsto (8).$$

Write $\tau(\alpha)$ for the first time at which the orbit becomes fixed.

For a target $\beta = (b_1, \ldots, b_k) \in \mathrm{Comp}(n)$, introduce the polynomial

$$\Phi_\beta(u) = \sum_{\substack{s_i \in \mathbb{Z}_{>0},\ s_i \mid b_i\ (1 \le i \le k) \\ s_i \neq s_{i+1}\ (1 \le i < k)}} u^{\sum_{i=1}^k b_i/s_i}. \tag{2}$$

The empty sum is zero.

Theorem 1 (clock and complete one-step inverse)**.** *For every $n \geq 1$, the following statements hold.*

(1) *The fixed points of A_n are exactly the compositions with adjacent parts unequal. Every orbit is absorbed at one of them.*

(2) *The sharp absorption clock is*

$$\max_{\alpha \in \mathrm{Comp}(n)} \tau(\alpha) = \lfloor \log_2 n \rfloor. \tag{3}$$

An explicit equality witness exists for every n.

(3) *For every target $\beta \in \mathrm{Comp}(n)$ and every $\ell \geq 1$,*

$$[u^\ell]\Phi_\beta(u) = \#\{\alpha \in A_n^{-1}(\beta) : \alpha \text{ has } \ell \text{ parts}\}. \tag{4}$$

In particular, $|A_n^{-1}(\beta)| = \Phi_\beta(1)$, and β lies in the image precisely when Φ_β is nonzero.

The fixed class in part (1) is the classical class of Carlitz compositions [5]. Runs of equal parts, their bounded lengths, and their asymptotics have been studied in considerably greater generality [1, 3, 6]. None of those static results is claimed here. The object of the note is the iterated, weight-preserving map (1), its sharp all-size clock, and its target-resolved inverse.

2. Termination and the sharp clock

The total weight is visibly invariant under A_n. If a state is not fixed, at least one run has length at least two, and replacing all runs decreases the number of parts strictly. Thus every orbit terminates. A sharper argument uses a backwards chain of newly created equalities.

Lemma 2 (doubling ancestry)**.** *If an orbit performs t nonfixed updates, then its total weight is at least 2^t.*

Proof. Put $\alpha^{(j)} = A_n^j(\alpha)$ for $0 \le j \le t$. There is nothing to prove when $t = 0$, so suppose $t \ge 1$. We first isolate the one-step ancestry claim used below. If a nontrivial maximal run in $\alpha^{(j)}$, with $j \ge 1$, is collapsed at the next update, choose two adjacent parts of that run. They are adjacent outputs of two consecutive maximal runs of $\alpha^{(j-1)}$. Those two preceding runs cannot both be singletons: if they were, their unchanged output values would be equal and the two runs would instead form one maximal run. Thus at least one chosen input part was itself produced by a nontrivial collapse in the preceding update.

Choose a nontrivial run in $\alpha^{(t-1)}$ and call its output in $\alpha^{(t)}$ the part w_t. When $t \ge 2$, apply the ancestry claim to choose among its equal inputs a part w_{t-1} that was produced nontrivially from $\alpha^{(t-2)}$; repeat down to a nontrivially produced part w_1 of $\alpha^{(1)}$. For $2 \le j \le t$, the run producing w_j has at least two inputs, all equal to the selected input w_{j-1}, and hence $w_j \ge 2w_{j-1}$. The first selected output satisfies $w_1 \ge 2$ because it combines at least two positive original parts. Therefore

$$2^t \le w_t \le n,$$

the last inequality holding because w_t is one part of a composition of total weight n. □

Lemma 2 gives $\tau(\alpha) \le \lfloor \log_2 n \rfloor$. It remains to attain the bound for every n, including values between powers of two.

For $t \ge 1$ put

$$C_t = (1, 1, 2, 4, \dots, 2^{t-1}), \tag{5}$$

which has weight 2^t. After $j < t$ updates, its active prefix is $(2^j, 2^j)$, followed by $2^{j+1}, \dots, 2^{t-1}$; hence it takes exactly t updates to reach (2^t).

Now let $t = \lfloor \log_2 n \rfloor$ and $r = n - 2^t$. For $n = 1$ use (1). For $n > 1$, define

$$W_n = \begin{cases} C_t, & r = 0, \\ (r, C_t), & r = 2^{t-1}, \\ (C_t, r), & 0 < r < 2^t,\ r \ne 2^{t-1}. \end{cases} \tag{6}$$

If $0 < r < 2^t$ and $r \ne 2^{t-1}$, then for $0 \le j < t$ the orbit is

$$A_n^j(W_n) = (2^j, 2^j, 2^{j+1}, \dots, 2^{t-1}, r),$$

where the power string after the repeated pair is empty when $j = t - 1$. Indeed, the appended r never equals its unchanged left neighbour 2^{t-1}. The tth state is $(2^t, r)$, which is fixed.

If $r = 2^{t-1}$ and $t \ge 2$, then for $0 \le j < t - 1$,

$$A_n^j(W_n) = (r, 2^j, 2^j, 2^{j+1}, \dots, 2^{t-1}),$$

and $A_n^{t-1}(W_n) = (r, r, r)$, followed by the fixed state $(3r)$. The remaining bases are explicit: $W_1 = (1)$ has depth zero, $W_2 = C_1 = (1, 1)$ has depth one, and the half-remainder case $W_3 = (1, 1, 1)$ also has depth one. Thus W_n has depth t in every case. This proves (3) and the equality statement.

3. The divisor-path inverse

Proof of Theorem 1(3). Let α map to $\beta = (b_1, \dots, b_k)$. The unique maximal run of α producing b_i has some common value s_i and length b_i/s_i; therefore $s_i \mid b_i$. Consecutive runs are maximal, so $s_i \ne s_{i+1}$. The source length is $\sum_i b_i/s_i$.

Conversely, choose a divisor $s_i \mid b_i$ for every i, with adjacent choices unequal, and expand b_i into b_i/s_i copies of s_i. The adjacent inequality makes these expanded blocks precisely the maximal runs, so the expansion maps to β. The two constructions are inverse. Sorting the choices by total expanded length proves (4) and all its consequences. □

Formula (2) is also an efficient exact atlas. At layer i one keeps the last divisor s_i and a polynomial in the accumulated source length; transitions are allowed between unequal divisors. Thus every target fibre can be computed by a weighted path dynamic program over its divisor sets, without enumerating all 2^{n-1} source compositions.

For completeness, if $C(x)$ includes the empty fixed composition, the classical fixed-class generating function follows from a one-line last-part decomposition. If $C_j(x)$ counts nonempty fixed compositions ending in j, then $C_j = x^j(C - C_j)$, whence

$$C(x) = \frac{1}{1 - \sum_{j \geq 1} \frac{x^j}{1 + x^j}}. \tag{7}$$

Equation (7) is included only to identify the recurrent class; it receives no contribution credit.

4. Exact falsification and boundary

The accompanying standard-library program enumerates every positive composition of every total $1 \leq n \leq 18$. In a fresh process it checks total preservation, strict descent, absence of nontrivial cycles, every endpoint, the pointwise logarithmic bound, the witness (6), the classical fixed census, and (2) for every possible target, including empty fibres. The computation covers 262,143 states and executes 2,690,869 exact assertions. For $n = 18$ the depth profile is

$$10843 + 60946z + 47626z^2 + 11415z^3 + 242z^4.$$

These finite checks are counterexample pressure and reproducibility evidence, not proofs of the all-parameter statements.

The closest occupied internal carrier is a composition-refinement system, but that map splits balanced blocks and uses a suffix decoder. Here the operation coarsens maximal equal runs, the clock is forced by weight-doubling ancestry, and the inverse is arithmetic on adjacent divisor sets. Ordinary run-length encoding likewise records a value/count pair and changes representation; A_n multiplies the pair, preserves total weight, and iterates inside the same finite carrier. Recent work on random weak-composition growth studies the evolution and disappearance of adjacent equalities [2], while recent Arndt–Carlitz work remains a static adjacent-restriction study [4]; both receive zero contribution credit and use different objects. A bounded source search found no direct owner for this exact conjunction, but a search non-hit is not a novelty or priority certificate.

Data availability

No external data were used. The exact verifier, frozen transcript, and claim ledger are included with this manuscript.

Ethics declaration

This mathematical study involved no human participants, animals, personal data, or field intervention; ethics approval was not applicable.

Author contributions

For anonymous internal review, a single anonymous author is recorded as responsible for conceptualization, formal analysis, software, validation, writing, and reproducibility curation.

Conflict of interest

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Limitations and external status

The owner search was bounded and terminology dependent. The paper does not claim novelty for Carlitz compositions or run statistics, and it does not analyze random initial states, asymptotic depth distributions, or fibres of higher iterates. All public posting, submission, priority, authorship, and release actions remain under `HOLD_EXTERNAL`.

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